AI-Based Fertilizer Recommendation System Using Soil and Crop Data: A Machine Learning Approach for Precision Agriculture

Name of Student:

Department:

Registration No.:

Name of Research Supervisor:

Abstract

In the traditional agricultural landscape, fertilizers are applied on a trial and error basis or following general advice, resulting in inefficient fertilizer use, environmental degradation and lower crop yields. This research tries to overcome these challenges by designing and testing an AI-based fertiliser recommendation system based on machine learning for precision agriculture. The system requires a full set of spatially variable soil and crop data (pH, temperature, soil moisture, nutrient levels N, P, K, type of crop, past yield). In this study, the problem is modeled as a supervised learning problem, which is a joint classification problem and regression problem.

Five ML models (namely, Random Forest, Decision Tree, Logistic Regression, K-Nearest neighbour (KNN), and XGBoost) are trained and tested. As for the results, an ensemble, specifically XGBoost, has performed better than other simpler models with an accuracy and an F1 Score of 0.6241. Finally, explanatory notes are added in the final model for reasons of usability for the user and trust by the user. In conclusion, this research provides a practical, data-driven solution that can help farmers optimize the use of inputs, increase productivity, and contribute to more sustainable agricultural practices.

Acknowledgment

I would like to extend my deepest gratitude to my research supervisor for their invaluable guidance, insight, and patience throughout this project. This dissertation would not have been possible without their continuous support and expert advice. I am also grateful to the University of East London for providing the resources and academic environment necessary for this research. A special thanks to my family and friends for their unwavering encouragement and support during this academic journey.

List of Acronyms

Acronym	Full Form
AI	Artificial Intelligence
ANN	Artificial Neural Networks
GBM	Gradient Boosting Machines
GIS	Geographic Information Systems
ICAR	Indian Council of Agricultural Research
IoT	Internet of Things
KNN	K-Nearest Neighbors
LIME	Local Interpretable Model-agnostic Explanations
MAE	Mean Absolute Error
MLP	Multi-Layer Perceptrons
ML	Machine Learning
MSE	Mean Squared Error
N	Nitrogen
P	Phosphorus
K	Potassium
SVM	Support Vector Machines
SHAP	SHapley Additive exPlanations
UAV	Unma

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Chapter 1: Introduction

Long methods of farming are usually based on guesswork or universal technique of using fertilizers causing overuse, soil wearing down and unfair crop production. The proposed research will deal with this problem by designing an AI-powered fertilizer recommendation system that would use soil and crop-specific information to make farming more precise through AI (Musanase et al., 2023). Under machine learning, the system will auto-analyze factors and parameters that include: the soil pH, nitrogen, phosphorus, potassium, moisture, and temperature as well as the crop type and its previous yield data to make a recommendation on the type of fertilizer and the amount required. The solution proposed will be dynamic and constantly learn and adapt to the regional requirements, unlike the conventional static models (Xing and Wang, 2024). The method maximises the use of the fertiliser, lessens the effect on the environment, and also makes farming more productive. A different range of classification and regression algorithms will be tried and compared to create a recommendation engine, the effectiveness of which will be reviewed by means of accuracy, adaptability, and applicability in the real world.

1.1 Aim and Objectives

1.1.1 Aim

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Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
Supervisor Allocation												
Library session												
Milestone 1:- Topic selection												
Proposal preparation and review												
Milestone 2:- Proposal Submission												
Abstarct												
Chapter 1- Introduction												
Chapter 2- Literature Review												
Milestone 3:- Practical implementation/ Model Development												
Chapter 3- Methodology												
Chapter 4- Experimental Results Documentation												
Chapter 5- Conclusion and Future work												
Final review and Modifications												
Milestone 4:- Final Submission												

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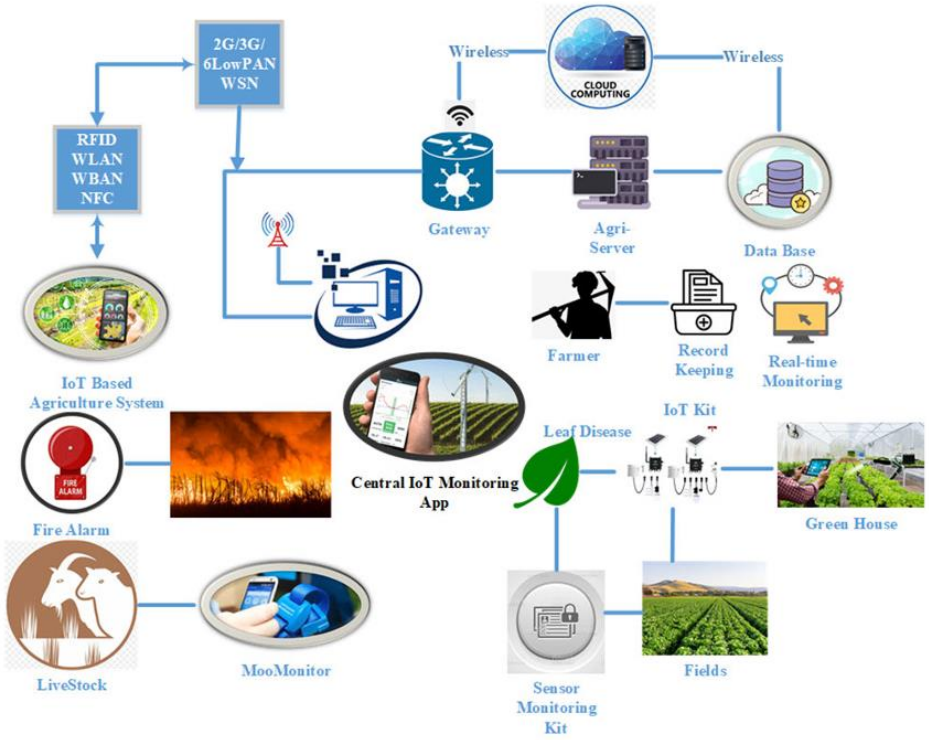

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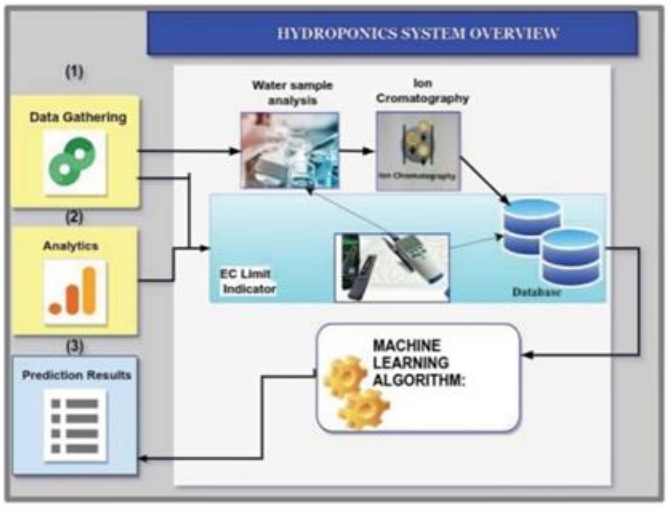

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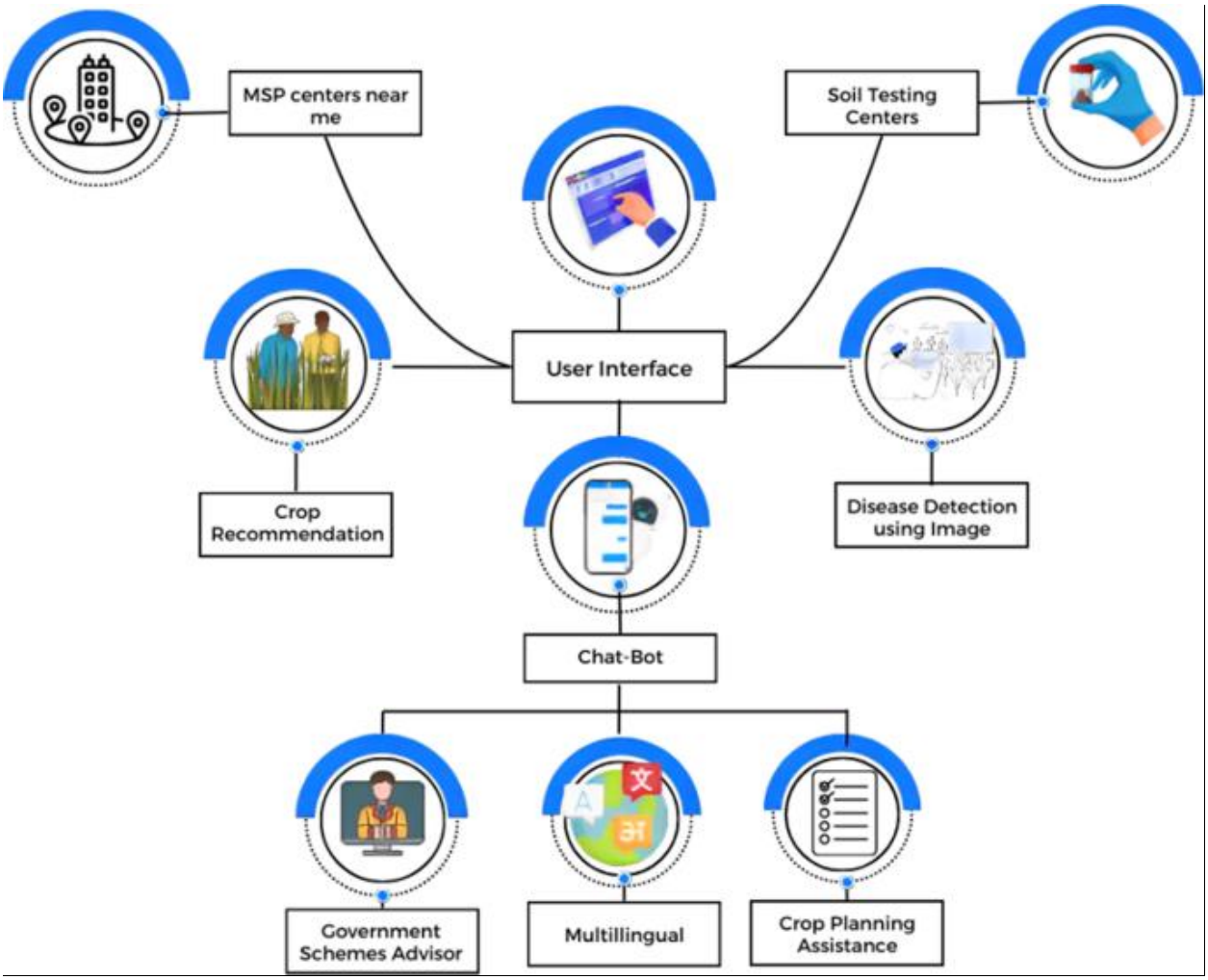

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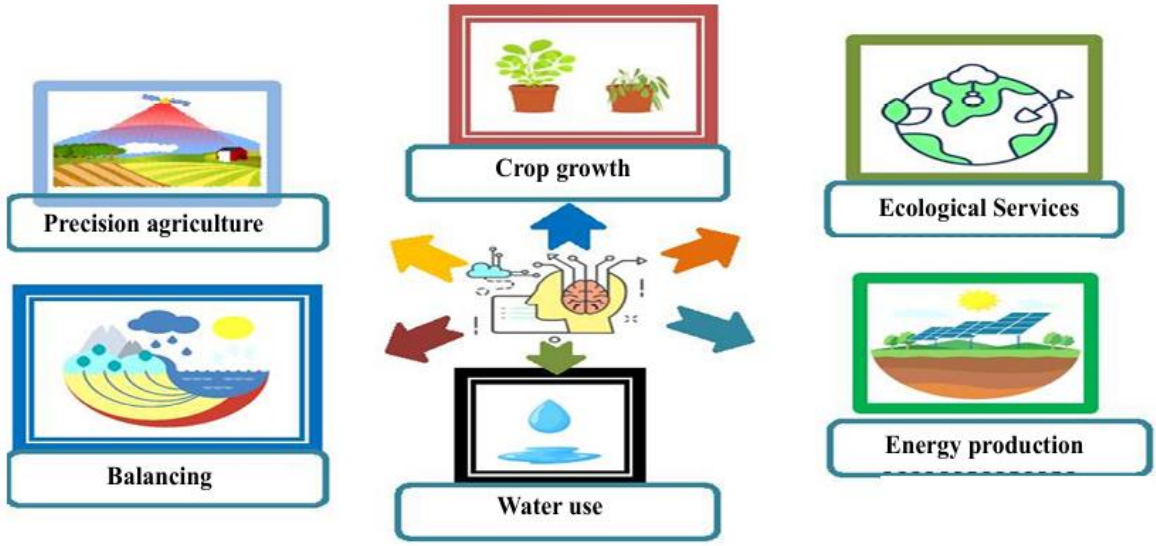
Crop growth
Precision agriculture
Ecological Services
Balancing
Energy production
Water use

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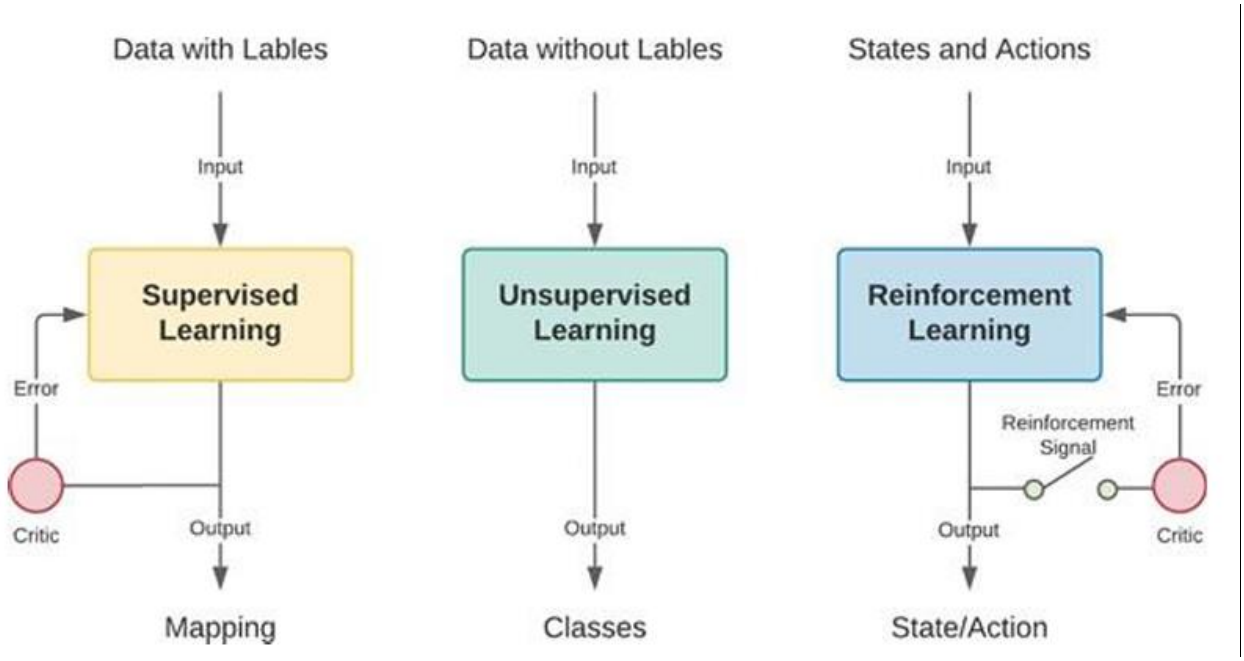

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Comparison Table: AI-Based Fertilizer Recommendation

Author and Year	Technique	Specification	Limitation	Dataset	Results
Khan et al. (2022)	IoT-assisted Context-Aware System	Sensor-driven crop and soil data for fertilizer	Limited scalability in low-connectivity areas	Custom IoT Deployment	Fertilizer use ↑ ~21%
Thorat et al. (2023)	TPF-CNN (Twin Parallel Fusion CNN)	Deep learning for fertilizer and insecticide recommendation	High computational cost	Smart Farm Sensor Dataset	Accuracy: 96.3%
Latha and Kumaresan (2025)	Deep Learning and Soil Nutrient Prediction	DL with strategic crop and soil nutrient recommendation	Needs high-quality soil data	Soil survey and lab data	Nutrient prediction: 93.8%
Awais et al. (2023)	ML for Soil Analysis	ML with soil feature engineering	Poor generalization across regions	Global soil datasets	Yield ↑ 15–18%
Chaima (2023)	IoT and ML Fertilization Model	Smart ML-based fertilization in arid zones	Sensor calibration and power issues	Arid zone IoT data	Efficiency ↑ ~20%
Soy and Prakash (2025)	Multispectral AI-based Mapping	Multispectral imaging for nutrient maps	Costly data acquisition (UAV/satellite)	UAV and satellite data	Mapping accuracy: 94.7%
Hossain and Islam (2022)	AI for Soil-Crop Suitability	AI evaluates soil-crop compatibility	Low tech adoption in rural areas	Bangladesh soil and climate data	Yield ↑ ~18%
Akintuyi (2024)	Adaptive AI Algorithms	Self-learning fertilizer optimization system	Real-time prediction errors	Sensor and cloud system	Self-optimized plans over time
Elbasi et al. (2024)	AI-Powered Decision-Making	Automated AI fertilizer pipelines	Needs clean labeled data	IoT and drone data	Decision time ↓ ~30%
Murugamani et al. (2022)	ML in 5G IoT-based Precision Agri	5G-enabled ML for real-time fertilizer management	5G not yet widely available	Smart farm and public datasets	Rec. precision: 92.5%

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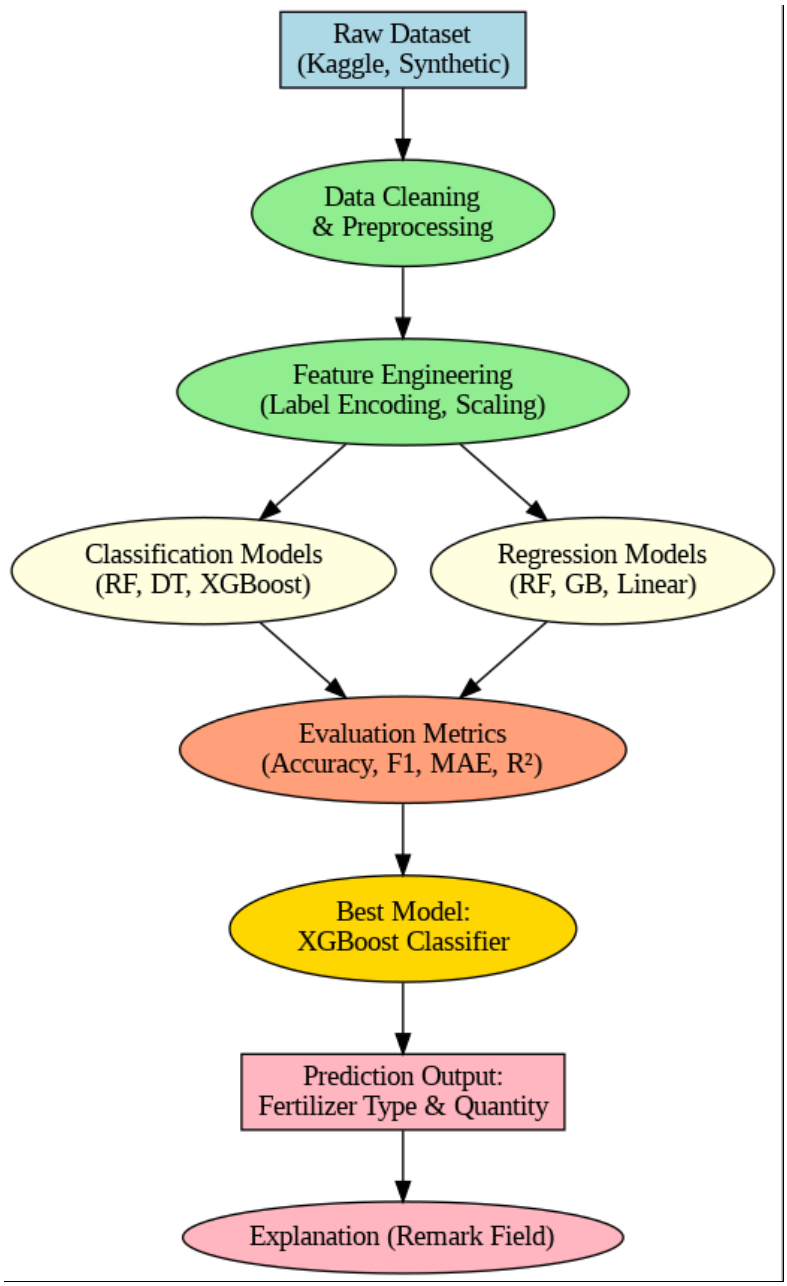
Raw Dataset
(Kaggle, Synthetic)
Data Cleaning
& Preprocessing
Feature Engineering
(Label Encoding, Scaling)
Classification Models
(RF, DT, XGBoost)
Regression Models
(RF, GB, Linear)
Evaluation Metrics
(Accuracy, F1, MAE, R²)
Best Model:
XGBoost Classifier
Prediction Output:
Fertilizer Type & Quantity
Explanation (Remark Field)

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```
Dataset has been loaded successfully
Number of rows and columns in the dataset: (3100, 12)
Here are the column names in the dataset:
['Temperature', 'Moisture', 'Rainfall', 'PH', 'Nitrogen', 'Phosphorous', 'Potassium', 'Carbon', 'Soil', 'Crop', 'Fertilizer', 'Remark']
```

	Temperature	Moisture	Rainfall	PH	Nitrogen	Phosphorous	Potassium	Carbon	Soil	Crop	Fertilizer	Remark
0	50.179845	0.725893	205.600816	6.227358	66.701872	76.963560	96.429065	0.496300	Loamy Soil	rice	Compost	Enhances organic matter and improves soil stru...
1	21.633318	0.721958	306.081601	7.173131	71.583316	163.057636	148.128347	1.234242	Loamy Soil	rice	Balanced NPK Fertilizer	Provides a balanced mix of nitrogen, phosphoru...
2	23.060964	0.685751	259.336414	7.380793	75.709830	62.091508	80.308971	1.795650	Peaty Soil	rice	Water Retaining Fertilizer	Improves water retention in dry soils. Prefer ...
3	26.241975	0.755095	212.703513	6.883367	78.033687	151.012521	153.005712	1.517556	Loamy Soil	rice	Balanced NPK Fertilizer	Provides a balanced mix of nitrogen, phosphoru...
4	21.490157	0.730672	268.786767	7.578760	71.765123	66.257371	97.000886	1.782985	Peaty Soil	rice	Organic Fertilizer	Enhances fertility naturally, ideal for peaty ...

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```
Checking for missing values in each column
Temperature    0
Moisture       0
Rainfall       0
PH             0
Nitrogen       0
Phosphorous    0
Potassium      0
Carbon         0
Soil           0
Crop           0
Fertilizer     0
Remark         0
dtype: int64
```

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```
Unique values in 'Soil': ['Loamy Soil' 'Peaty Soil' 'Acidic Soil' 'Neutral Soil' 'Alkaline Soil']
Unique values in 'Crop': ['rice' 'wheat' 'Mung Bean' 'Tea' 'millet' 'maize' 'Lentil' 'Jute'
 'Coffee' 'Cotton' 'Ground Nut' 'Peas' 'Rubber' 'Sugarcane' 'Tobacco'
 'Kidney Beans' 'Moth Beans' 'Coconut' 'Black gram' 'Adzuki Beans'
 'Pigeon Peas' 'Chickpea' 'banana' 'grapes' 'apple' 'mango' 'muskmelon'
 'orange' 'papaya' 'pomegranate' 'watermelon']
Unique values in 'Fertilizer': ['Compost' 'Balanced NPK Fertilizer' 'Water Retaining Fertilizer'
 'Organic Fertilizer' 'Gypsum' 'Lime' 'DAP' 'Urea' 'Muriate of Potash'
 'General Purpose Fertilizer']
Unique values in 'Remark': ['Enhances organic matter and improves soil structure. Prefer this for low-carbon soils to boost soil health naturally.'
 'Provides a balanced mix of nitrogen, phosphorus, and potassium for loamy soils. Prefer this for general-purpose fertilization in well-structured soils.'
 'Improves water retention in dry soils. Prefer this for soils with low moisture to prevent water stress in plants.'
 'Enhances fertility naturally, ideal for peaty soils. Prefer this to improve soil fertility while maintaining organic farming practices.'
 'Corrects alkaline soil, adds calcium and sulfur. Prefer this to lower soil pH and enhance soil structure.'
 'Neutralizes acidic soil and improves pH balance. Prefer this to correct low soil pH, improving nutrient availability.'
 'Rich in phosphorus, essential for root development. Prefer this for phosphorus-deficient soils to improve plant establishment.'
 'Provides high nitrogen, ideal for rapid leafy growth. Prefer this for nitrogen-deficient soils as it supports vegetative growth.'
 'High potassium content, improves fruit and flower quality. Prefer this for potassium-deficient soils to enhance crop productivity.'
 'Suitable for general use across various soil and crop types. Prefer this when no specific deficiency is detected.']
```

	Temperature	Moisture	Rainfall	PH	Nitrogen	Phosphorous	Potassium	Carbon	Soil	Crop	Fertilizer	Remark	Fertilizer_Label
0	50.179845	0.725893	205.600816	6.227358	66.701872	76.963560	96.429065	0.496300	2	28	Compost	Enhances organic matter and improves soil stru...	1
1	21.633318	0.721958	306.081601	7.173131	71.583316	163.057636	148.128347	1.234242	2	28	Balanced NPK Fertilizer	Provides a balanced mix of nitrogen, phosphoru...	0
2	23.060964	0.685751	259.336414	7.380793	75.709830	62.091508	80.308971	1.795650	4	28	Water Retaining Fertilizer	Improves water retention in dry soils. Prefer ...	9
3	26.241975	0.755095	212.703513	6.883367	78.033687	151.012521	153.005712	1.517556	2	28	Balanced NPK Fertilizer	Provides a balanced mix of nitrogen, phosphoru...	0
4	21.490157	0.730672	268.786767	7.578760	71.765123	66.257371	97.000886	1.782985	4	28	Organic Fertilizer	Enhances fertility naturally, ideal for peaty ...	7

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```
Random Forest: Accuracy = 0.6204, F1 = 0.5806
```

```
XGBoost: Accuracy = 0.6231, F1 = 0.6241
```

```
Logistic Regression: Accuracy = 0.5739, F1 = 0.5139
```

```
KNN: Accuracy = 0.5965, F1 = 0.5840
```

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```
XGBoost: Accuracy = 0.6231, F1 = 0.6241
```

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```
Best Model: Decision Tree
```

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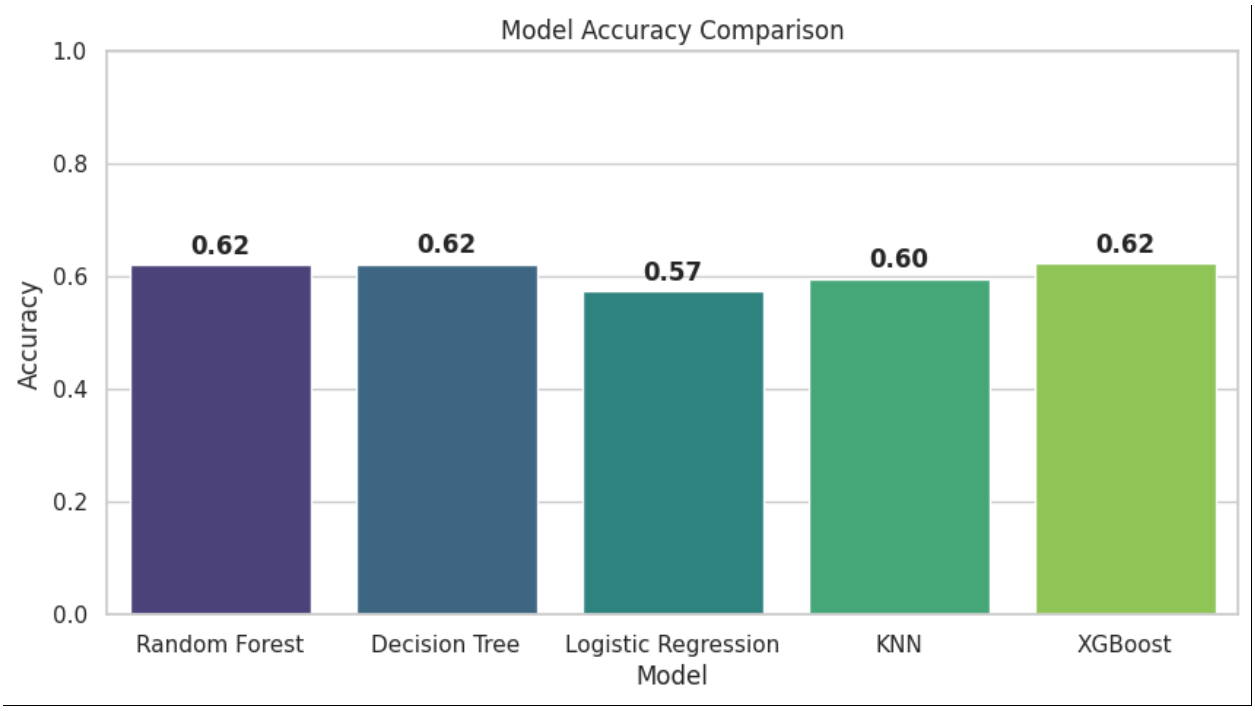

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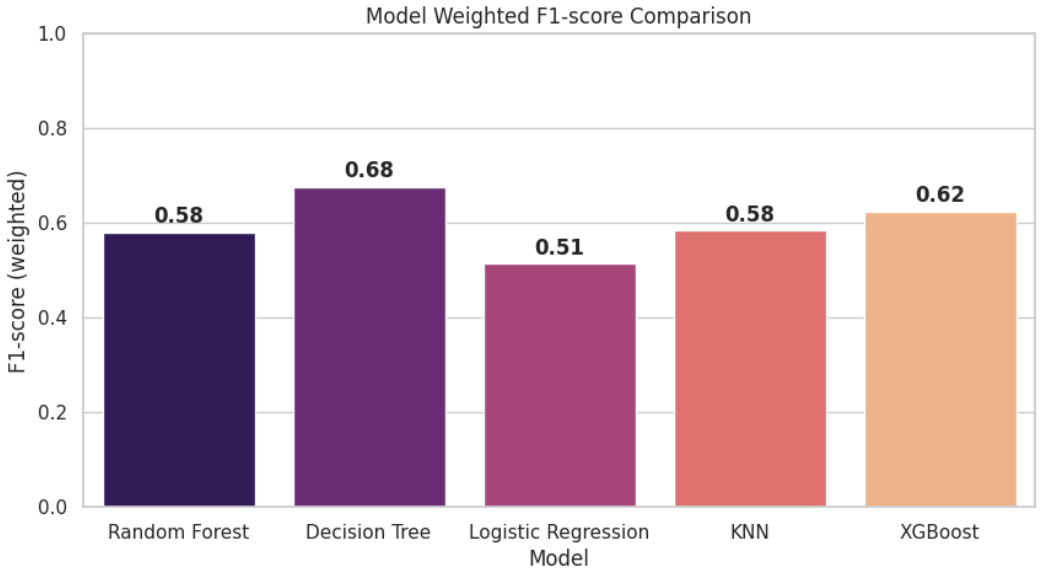

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```
[ ]  # First, import the pandas library to work with the dataset
     import pandas as pd

     # Specify the filename of the uploaded CSV file
     file_name = 'fertilizer_recommendation_dataset.csv'
     df = pd.read_csv(file_name)

     # Display confirmation message and the shape of the dataset
     print("Dataset has been loaded successfully")
     print("Number of rows and columns in the dataset:", df.shape)

     # Print the list of column names in the dataset
     print("Here are the column names in the dataset:")
     print(df.columns.tolist())

     # Display the first five rows to understand the structure and data
     df.head()
```

[redacted]

```
Checking data types of each column
Temperature    float64
Moisture       float64
Rainfall       float64
PH             float64
Nitrogen       float64
Phosphorous    float64
Potassium      float64
Carbon         float64
Soil            object
Crop            object
Fertilizer      object
Remark          object
dtype: object
```

```
print("\nChecking data types of each column")
print(df.dtypes)
```

```
[ ] from sklearn.preprocessing import LabelEncoder

    # Encode categorical columns
    le_soil = LabelEncoder()
    le_crop = LabelEncoder()
    le_fert = LabelEncoder()

    df['Soil'] = le_soil.fit_transform(df['Soil'])
    df['Crop'] = le_crop.fit_transform(df['Crop'])
    df['Fertilizer_Label'] = le_fert.fit_transform(df['Fertilizer'])  # New label column

    # Mapping dictionary to decode predictions later
    fertilizer_mapping = dict(zip(le_fert.transform(le_fert.classes_), le_fert.classes_))

    df.head()
```

	Temperature	Moisture	Rainfall	PH	Nitrogen	Phosphorous	Potassium	Carbon	Soil	Crop

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```
# feature selection

features = ['Temperature', 'Moisture', 'Rainfall', 'PH', 'Nitrogen',
            'Phosphorous', 'Potassium', 'Carbon', 'Soil', 'Crop']

X = df[features]
y = df['Fertilizer_Label']
```

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```
"Random Forest": RandomForestClassifier(max_depth=10, random_state=42),
```

```
"Decision Tree": DecisionTreeClassifier(max_depth=10, random_state=42),
```

```
"Logistic Regression": make_pipeline(StandardScaler(),LogisticRegression(max_iter=1000)),
```

```
"KNN": KNeighborsClassifier(),
```

```
"XGBoost": XGBClassifier(eval_metric='mlogloss')
```

```python
# Define classifiers
models = {
    "Random Forest": RandomForestClassifier(max_depth=10, random_state=42),
    "Decision Tree": DecisionTreeClassifier(max_depth=10, random_state=42),
    "Logistic Regression": make_pipeline(StandardScaler(),LogisticRegression(max_iter=1000)),
    "KNN": KNeighborsClassifier(),
    "XGBoost": XGBClassifier(eval_metric='mlogloss')
}

results = {}

for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    acc = accuracy_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred, average='weighted')
    results[name] = {"model": model, "accuracy": acc, "f1_score": f1}
    print(f"{name}: Accuracy = {acc:.4f}, F1 = {f1:.4f}")
```

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